

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, ML Evaluation & Experimentation Aug 2025 to May 2026

- Boosted malware-family classification to **80.2%** accuracy across **17 families** by soft-voting **ResNet50**, **ViT-B/16**, VGG16, HOG, and Haralick models, beating the best single model by **2.4 pts** (McNemar $p < 0.001$).
- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Ran **evals** on RAG workflows by comparing chunking strategies, embedding models, and top-k settings, and classified each failure as retrieval-caused or generation-caused.

Turito | Software Engineering Intern, Systems Performance Jun 2022 to Oct 2022

- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Prototyped a **Rust** + **QUIC** media path that supported about **8,000** lightweight subscribers with about **24%** lower startup time.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: C++17, Python, Rust, SQL

Evaluation & Stats: McNemar testing, controlled experiments, AUC/EER/TAR@FAR, Recall/NDCG, NumPy, Pandas, scikit-learn

ML: PyTorch, ViT, ResNet, ensembles, metric learning

Infra: Docker, gRPC, Kafka, telemetry pipelines, PostgreSQL, Linux, CMake

PROJECTS

Contactless Palmprint Verification | *PyTorch, OpenCV, Siamese networks*

- Fused EfficientNet, ResNet, ViT, LBP, KAZE, Gabor, DWT, and DCT features into a 256-d triplet-loss embedding reaching **0.9975 AUC** and **2.13% EER**.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnswlib and **2.5x** faster than FAISS on 100K vectors.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Advanced Algorithms, Distributed Computing, Biometric Security, Artificial Intelligence, Database Systems

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)